

PAPER_372 — Compressed UQFF with B/Bcrit Superconductivity

Star Magic UQFF Whitepaper Series

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Abstract

This paper presents the Compressed UQFF formulation, a multi-term master gravity equation that incorporates Newtonian gravity, Hubble expansion, superconductivity via the B/Bcrit flux-quenching factor, environmental coupling, cosmological constant contribution, quantum coherence, fluid dynamics, and dark matter perturbation. The framework is parameterised for seven astrophysical systems and validated via unit test against SGR1745-2900. This is the first UQFF formulation to explicitly incorporate Bardeen-Cooper-Schrieffer (BCS) superconductivity quenching into the gravitational coupling (Linear Meissner form; see PAPER_375 for the exponential form).

1. Master Equation

$$\begin{aligned} g_{\text{UQFF}}(r, t) = & \frac{GM(t)}{r^2} \cdot [1 + H(t, z)] \cdot \left[1 - \frac{B}{B_{\text{crit}}}\right] \cdot [1 + F_{\text{env}}] \\ & + (U_{g1} + U_{g2} + U'_{g3} + U_{g4}) + \frac{\Lambda c^2}{3} + \frac{\hbar}{\Delta x \cdot \Delta p} \int (\psi_{\text{total}}^* \hat{H} \psi_{\text{total}} dV) \cdot \frac{2\pi}{t_{\text{Hubble}}} \\ & + \rho_{\text{fluid}} V g + (M_{\text{vis}} + M_{\text{DM}}) \left(\frac{\delta \rho}{\rho} + \frac{3GM}{r^3} \right) \end{aligned}$$

where $H(t, z) = H_0 t$ (Newtonian cosmological expansion approximation), $H_0 = 2.269 \times 10^{-18} \text{ s}^{-1}$.

2. Modular Component Functions

Function	Formula	Constants
compressed_base	GM/r^2	$G = 6.674\text{e-}11$
compressed_expansion	$1 + H_0 t$	$H_0 = 2.269\text{e-}18 \text{ s}^{-1}$
compressed_super_adj	$1 - B/B_{\text{crit}}$	linear Meissner
compressed_env	1.0	default

Function	Formula	Constants
compressed_cosm	$\Lambda c^2/3$	$\Lambda = 1.1\text{e-}52 \text{ m}^{-2}$
compressed_quantum	$(\hbar/10^{-68}) \cdot 2.176 \times 10^{-18} \cdot (2\pi/t_H)$	$t_H = 4.35\text{e}17 \text{ s}$
compressed_fluid	$\rho_f V g_l$	from MUGESystem
compressed_perturbation	$(M + M_{DM})(\delta\rho/\rho + 3GM/r^3)$	$\delta\rho/\rho = 10^{-5}$

3. System Parameters

System	M (kg)	r (m)	B (T)	Bcrit (T)	Vsys (m ³)	ffluid (Hz)
Magnetar SGR1745-2900	2.984e30	1×10 ⁴	1×10 ¹⁰	1×10 ¹¹	4.189e12	1.269e-14
Sagittarius A*	8.155e36	1×10 ¹²	1×10 ⁻⁵	1×10 ⁻⁴	3.552e45	3.465e-8
Tapestry Star-birth	1.989e35	3.086e17	1×10 ⁻⁴	1×10 ⁻³	1×10 ⁵³	1×10 ⁻¹²
Westerlund 2	1.989e35	3.086e17	1×10 ⁻⁴	1×10 ⁻³	1×10 ⁵³	1×10 ⁻¹²
Pillars of Creation	1.989e32	9.46e15	1×10 ⁻⁴	1×10 ⁻³	3.552e48	8.457e-14
Rings of Relativity	1.989e36	3.086e17	1×10 ⁻⁵	1×10 ⁻⁴	1×10 ⁵⁴	1×10 ⁻⁹
Student's Guide Universe	1×10 ⁵³	1×10 ²⁶	1×10 ⁻¹⁰	1×10 ⁻⁹	1×10 ⁸⁰	1×10 ⁻¹⁸

4. Validation

Unit test: test_compute_compressed_MUGE(SGR1745-2900) - Expected: **1.782e39 m/s²** - (Dominated by compressed_base × expansion; B/Bcrit = 0.1 → 90% retention)

5. Physical Interpretation

The $[1 - B/B_{\text{crit}}]$ factor models the Meissner effect: as the magnetic field B approaches the critical field B_{crit} , gravitational coupling is quenched—mirroring how a Type-I superconductor expels magnetic flux below B_{crit} . The compressed UQFF thus unifies gravitomagnetic quenching with cosmological expansion and quantum corrections in a single framework. (For Type-II exponential treatment, see PAPER_375.)

6. Implementation

C++: STAR_MAGIC_09SEPT_UQFF_MODULE.cpp, namespace CompressedUQFF **Python:** CondensedPhysics4.py, class CompressedUQFFBcritSuperconductivityCalculator (CP4 #20) **WOLFRAM_TERM:** WOLFRAM_TERM_COMPRESSED_UQFF_BCRIT

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